PHILIPPINE SOCIETY OF
ANIMAL NUTRITIONISTS

39th ANNUAL CONVENTION

Fueling Progress: Shaping Animal Nutrition for a Competitive and Sustainable Future

6 OCTOBER 2026
Grand Ballroom,
Okada Manila









PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS
39th Annual Convention | October 6, 2026

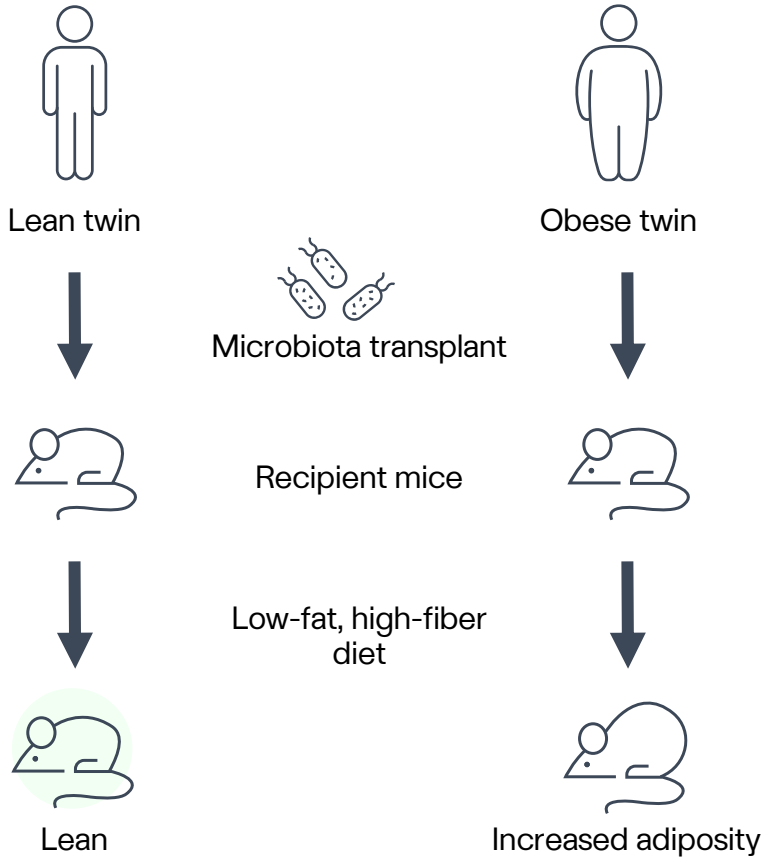
From Gut Microbiota to Host Resilience:

Integrating the Gut-Immune Axis for Sustainable Pig Production

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Regional Technical Manager Asia Pacific

Human Microbes Can Influence the Body Weight of Mice

- Microbiome from human twins with different weights
- Microbiome transplanted to similar germ-free mice
- Recipient mice developed similar phenotype as donor



Ridaura, Vanessa K., et al. "Gut microbiota from twins discordant for obesity modulate metabolism in mice." Science 341.6150 (2013): 1241214.

Childhood Asthma: From Interesting Associations to Deeper Understanding

- Several factors influence how well early life microbiome maturates
- Samples at 1 year of age predicts increased risk of asthma much later in life



Perinatal period

Mode of delivery

Antibiotic use in pregnancy

Diet during pregnancy

Environmental Exposures



Early life

Composition of the early microbiome

Disease Mediator



Childhood

Asthma Allergy

Disease Phenotypes

https://www.tandfonline.com/doi/full/10.1080/19490976.2025.2457489

The European sow microbiome

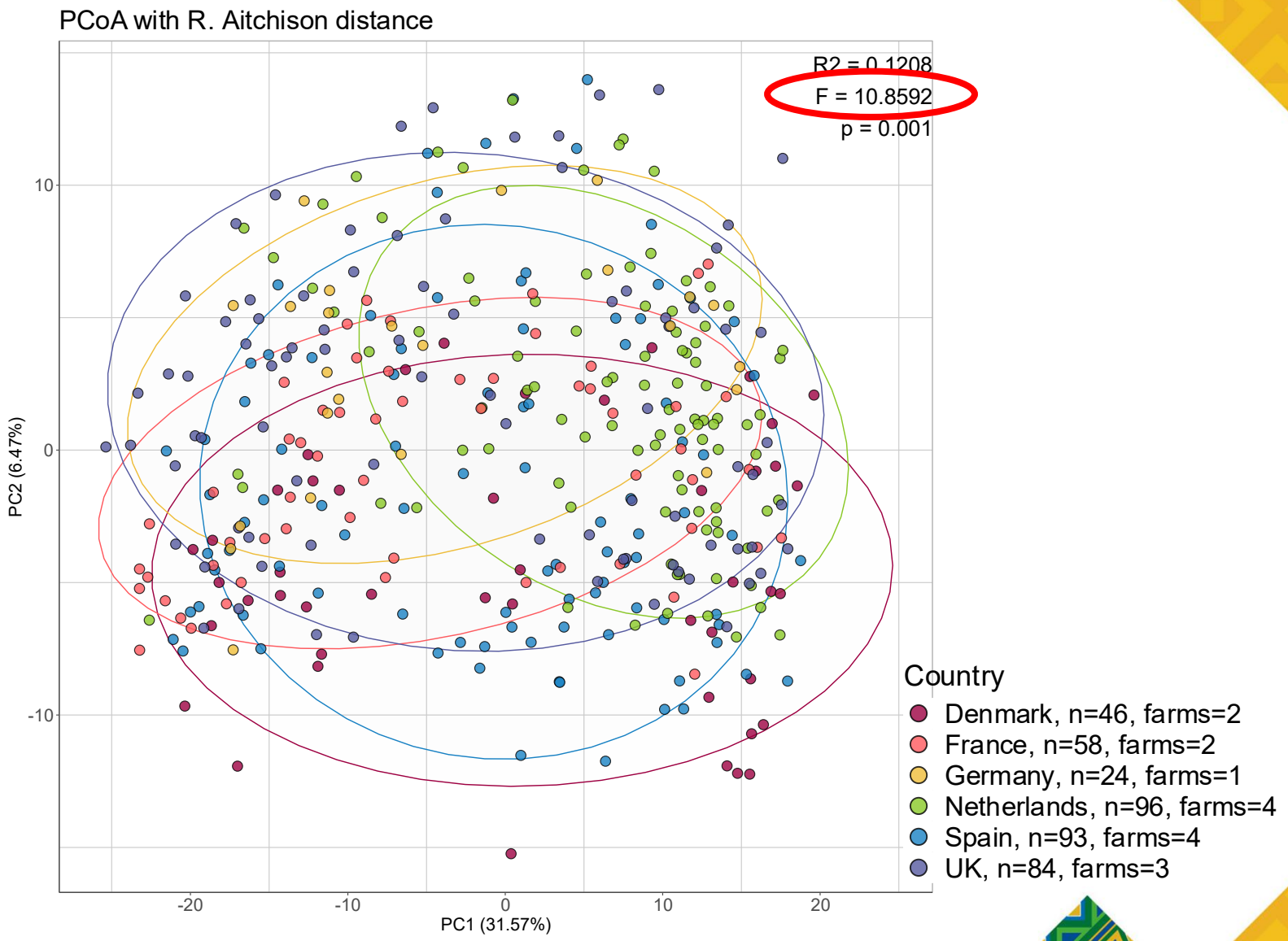


+666 samples

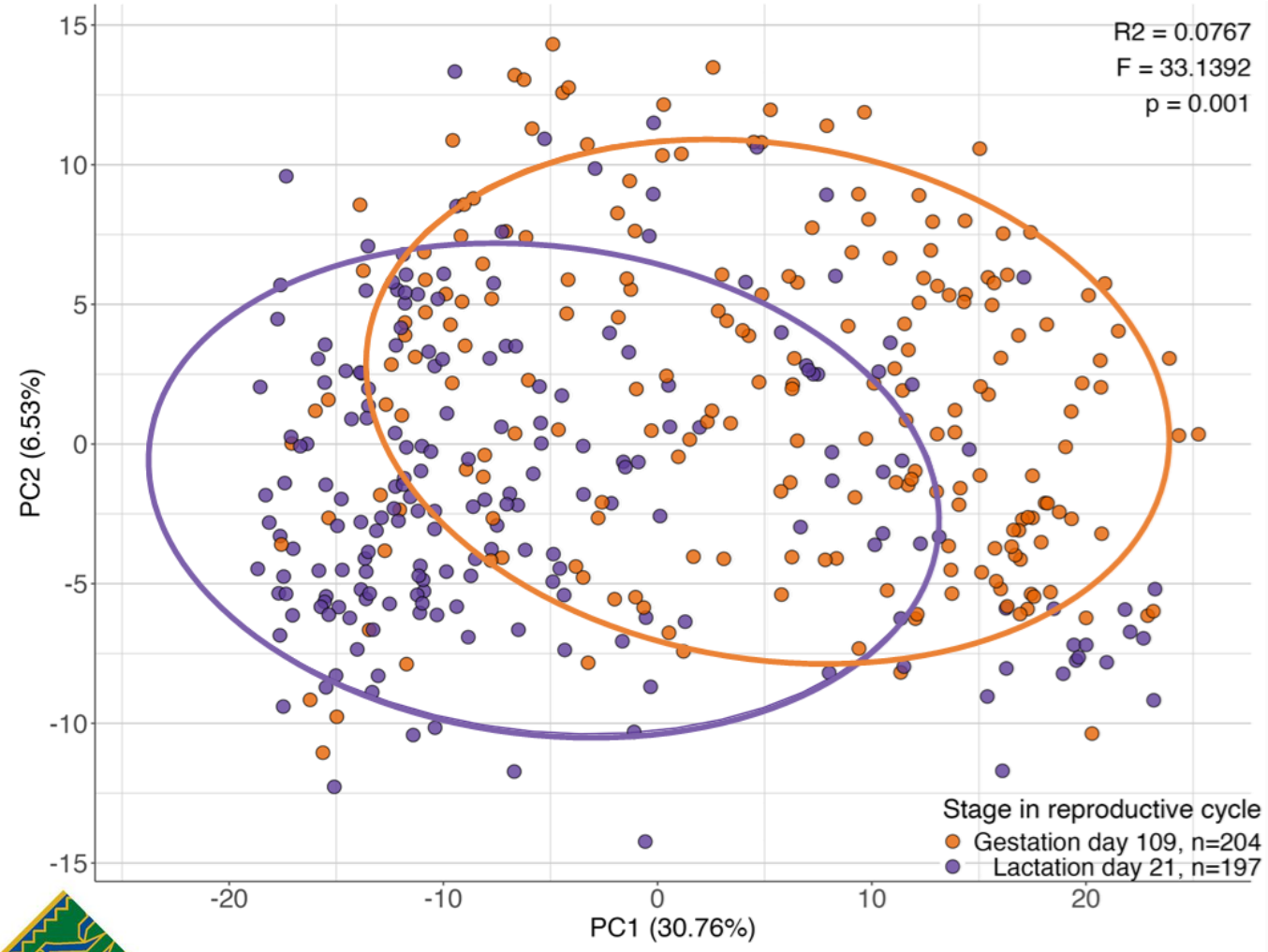
8 farms | 13 farms | 4 farms | 4 farms | 4 farms | 2 farms | 1 farm | 2 farms

Country effect

All countries are significantly distinct from each other tested with pairwise ANOVA
(P < 0.01)



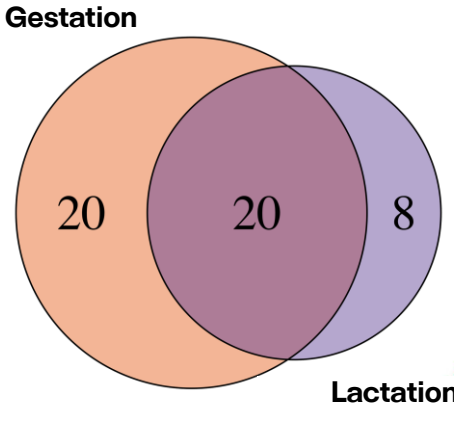
A shared microbiome



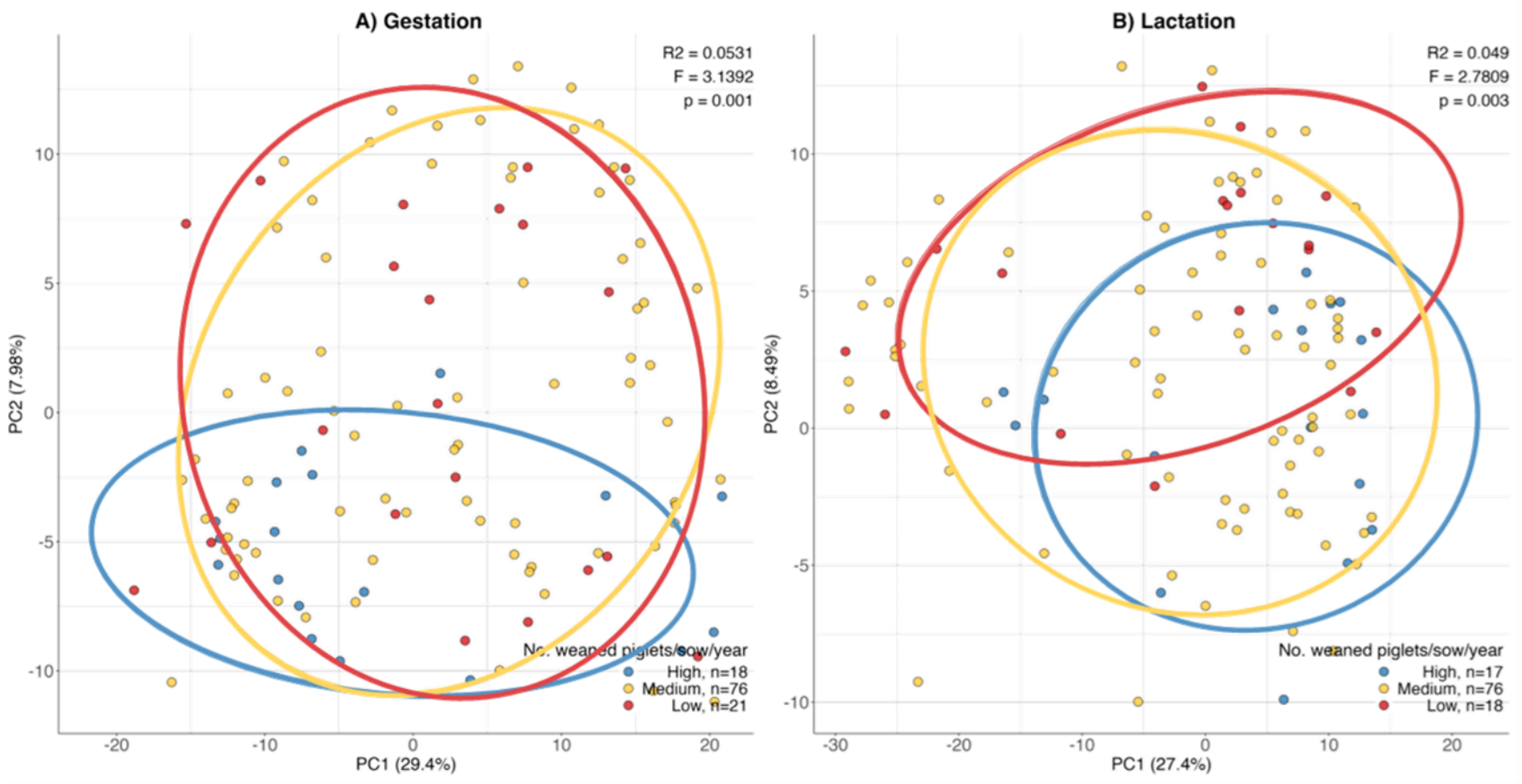
PCoA on robust Aitchinson distances

The biggest factor influencing the microbial composition is the time in the reproductive cycle.

Indicates a distinct gestation and a distinct lactation microbiome.



Link between sow gut microbiota and performance



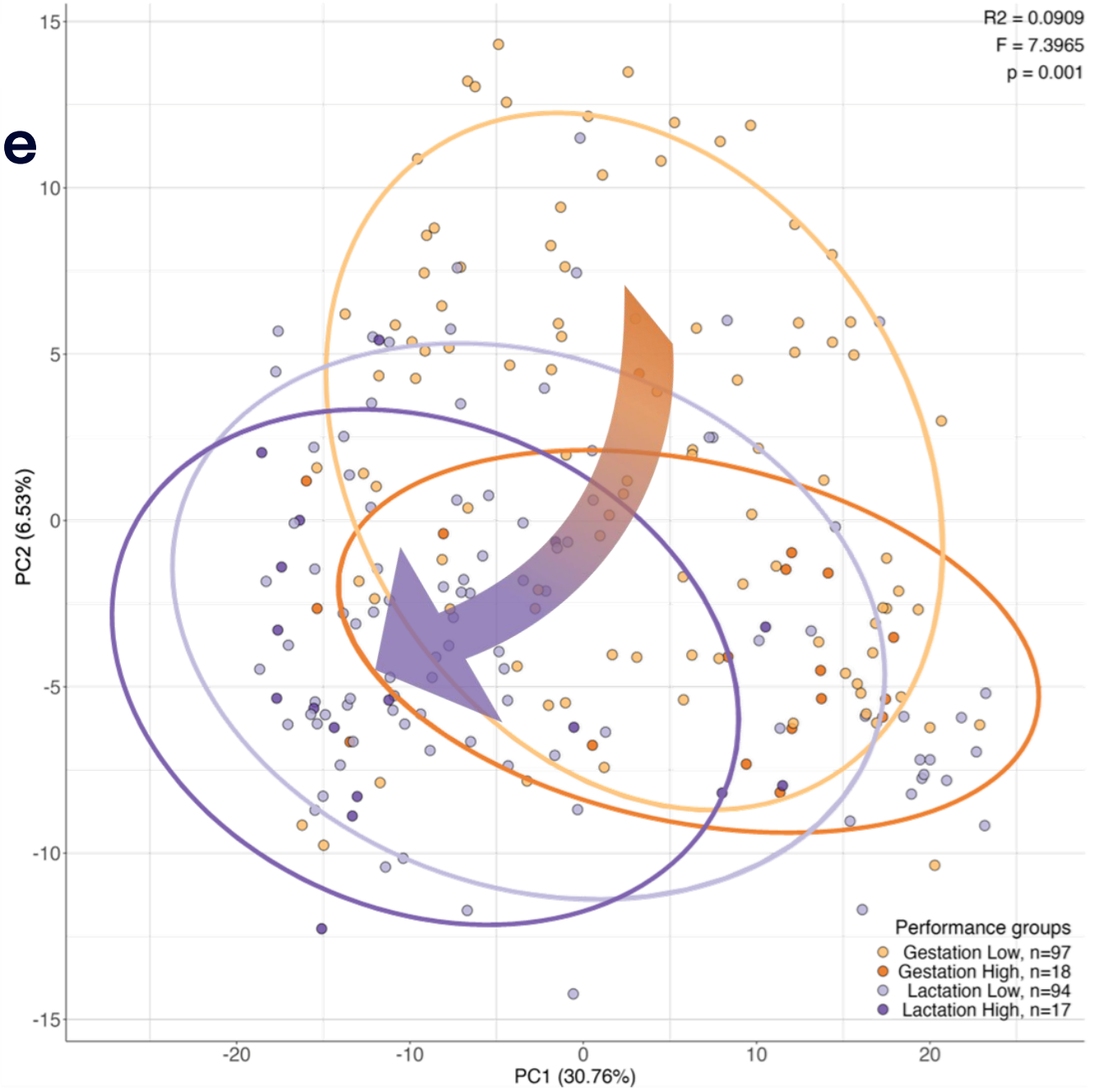
Significant differences between performance groups

– Mostly between High vs Low and High vs Medium

PCoA on robust Aitchinson distances

Performance groups across the two time points

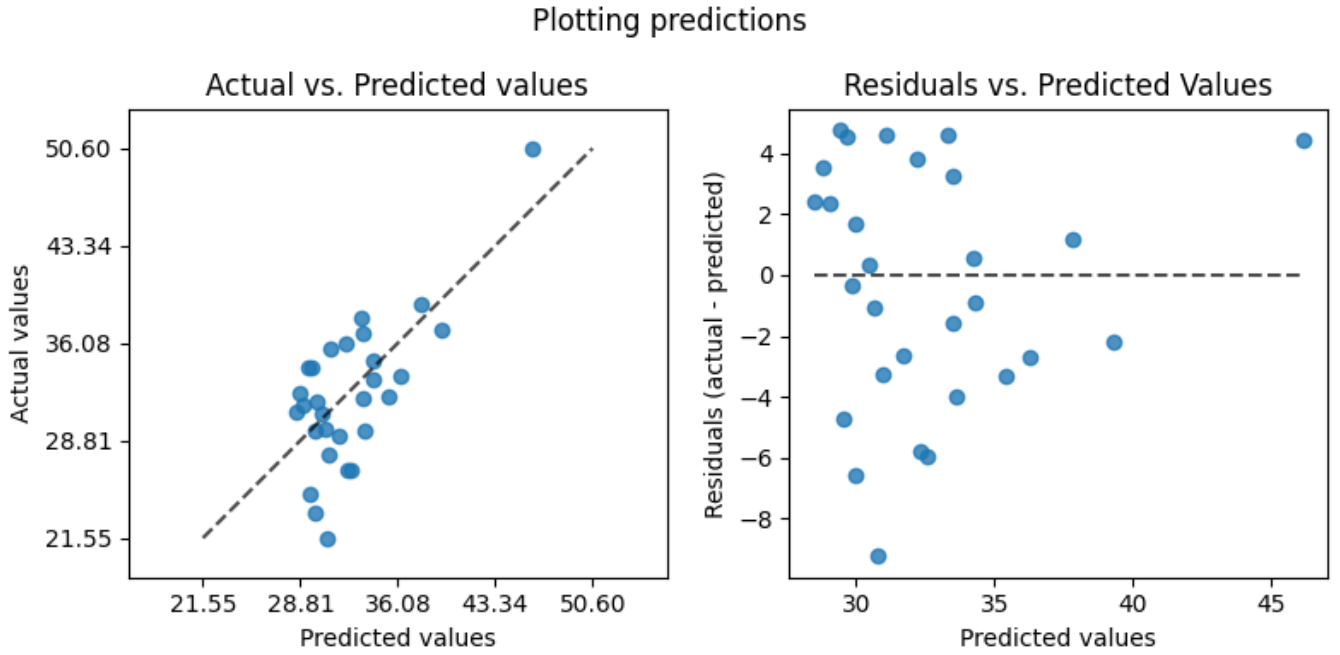
- Significant differences between all groups
- Lactation of high performers are furthest away from the gestation groups



Predicting productivity from lactation samples

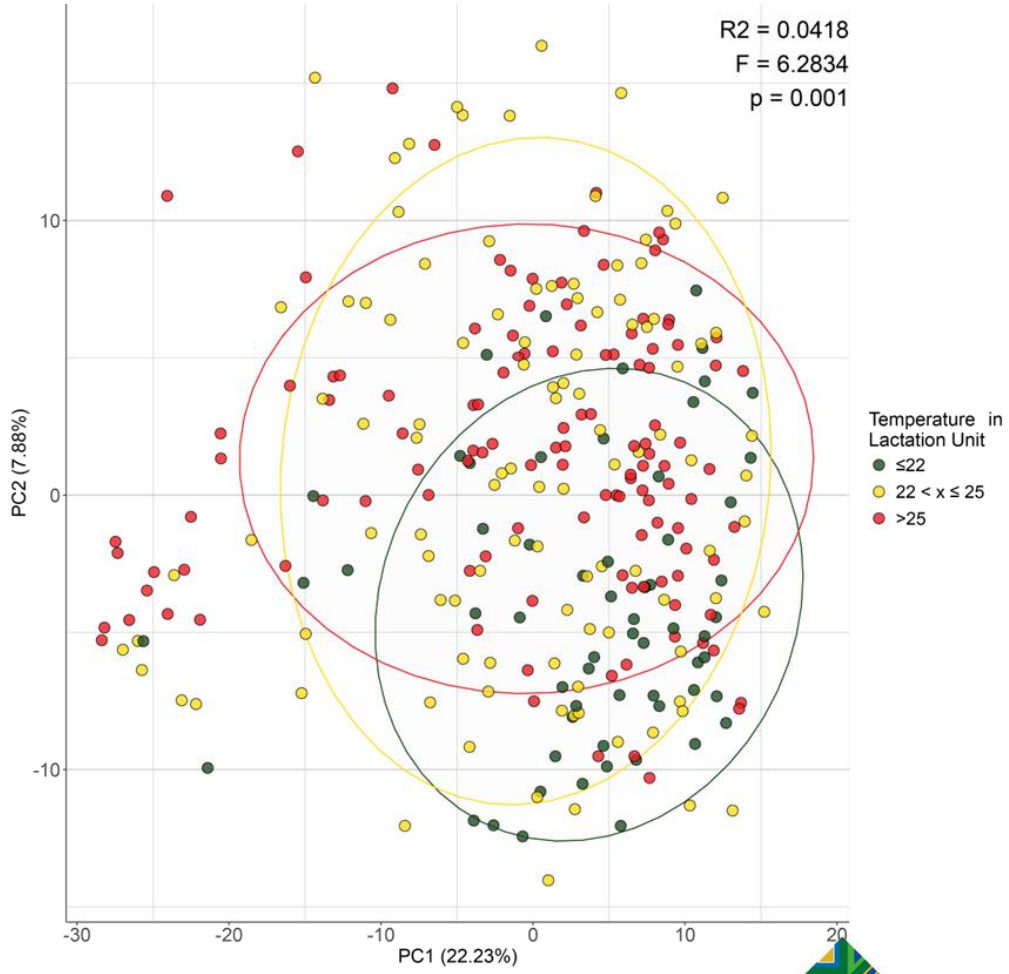
- From just 137 samples we can predict number of weaned piglets per sow per year with a Pearson correlation of 0.71

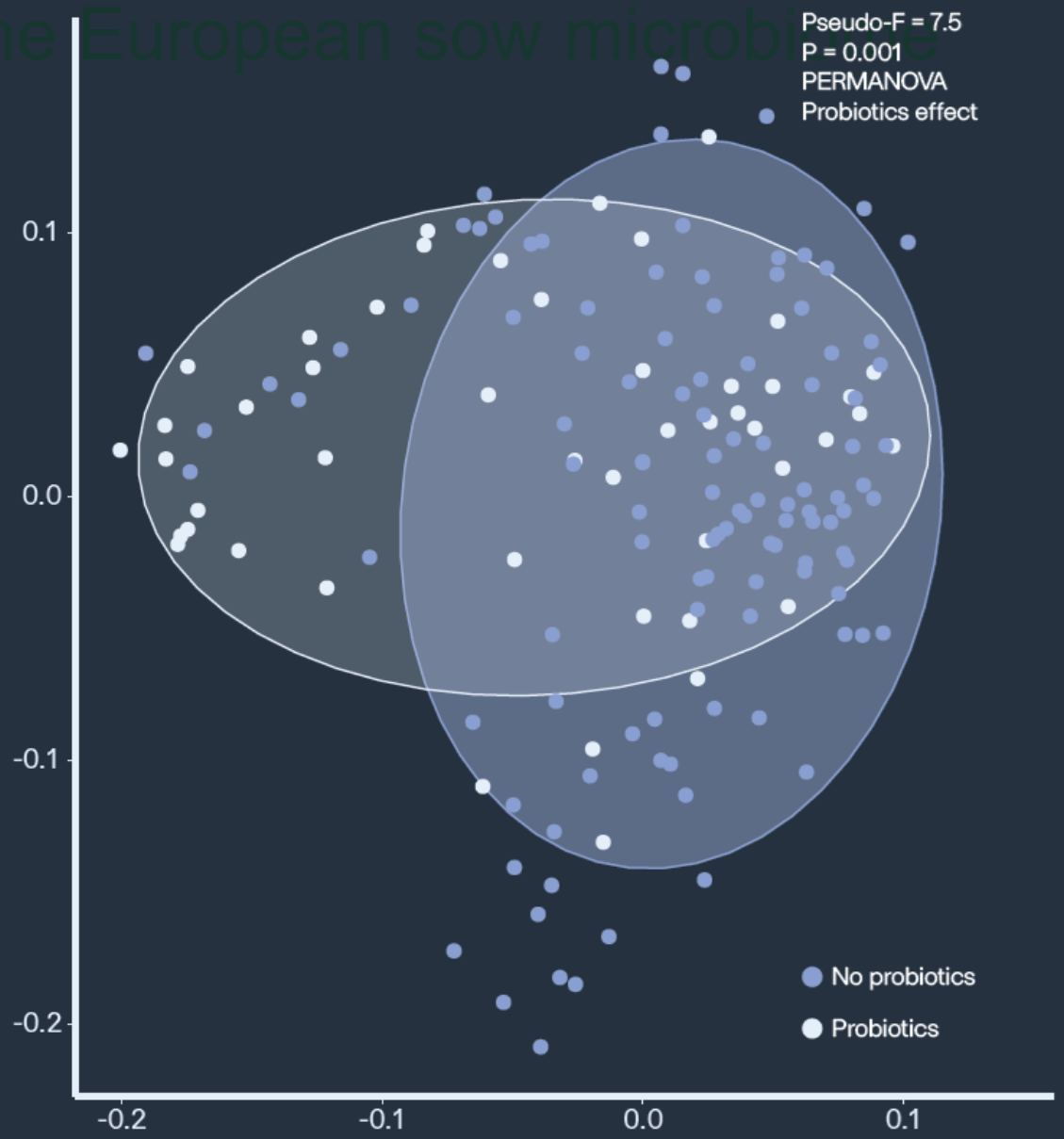
Interval coefficient	Relationship level
0.80-1.000	Very strong
0.60 – 0.799	Strong
0.40 – 0.599	Moderate
0.20–0.399	Weak
0.00–0.199	Very weak



From prediction to actionable management tool

- Actionable microbiome signatures such as farrowing stable temperature, Feed form, Floor Type etc.
- We identify significant effects linked to productivity such as farrowing stable temperature in the microbiome
- Identifying more signatures will allow us to provide actionable insights to our customers



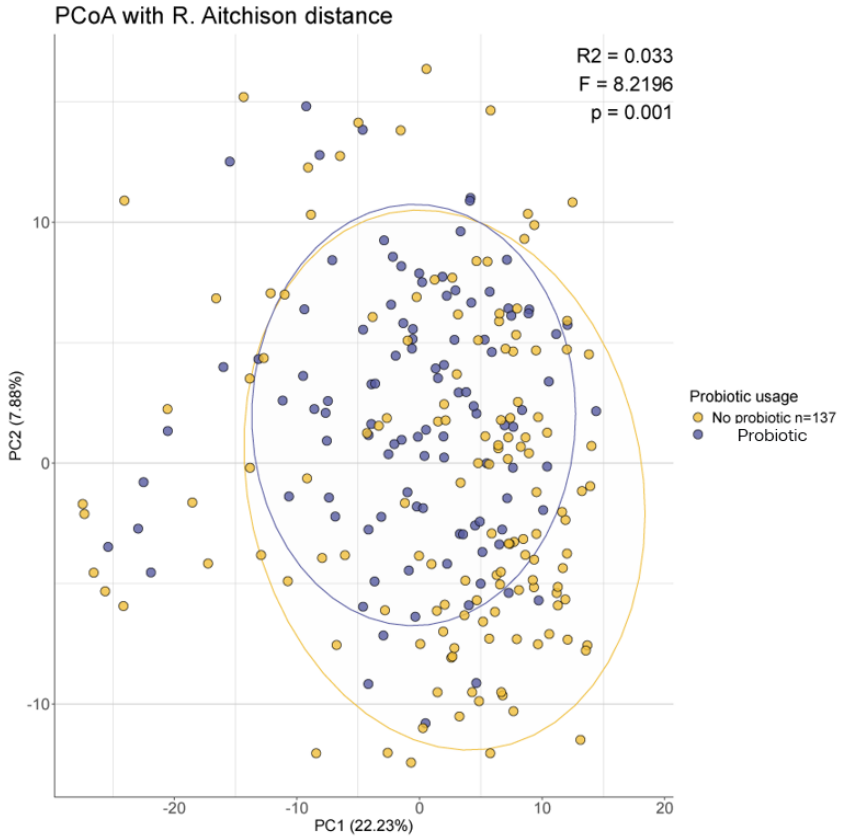


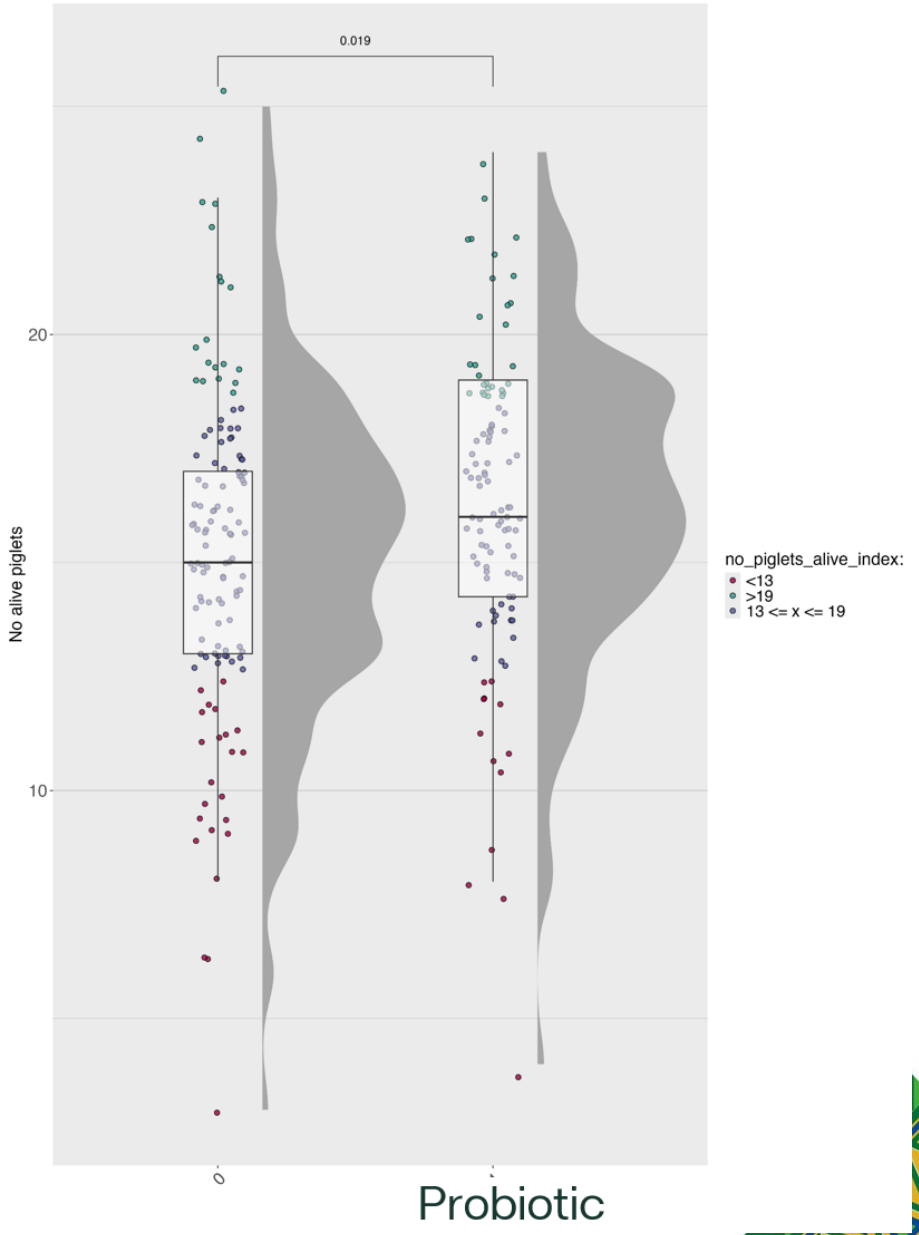
Probiotics

Modulate the Microbiome

Microbiome Modulated

- Significantly more live born piglets
- Significantly modulated microbiome





Why Gut Health Matters in Modern Pig Production?

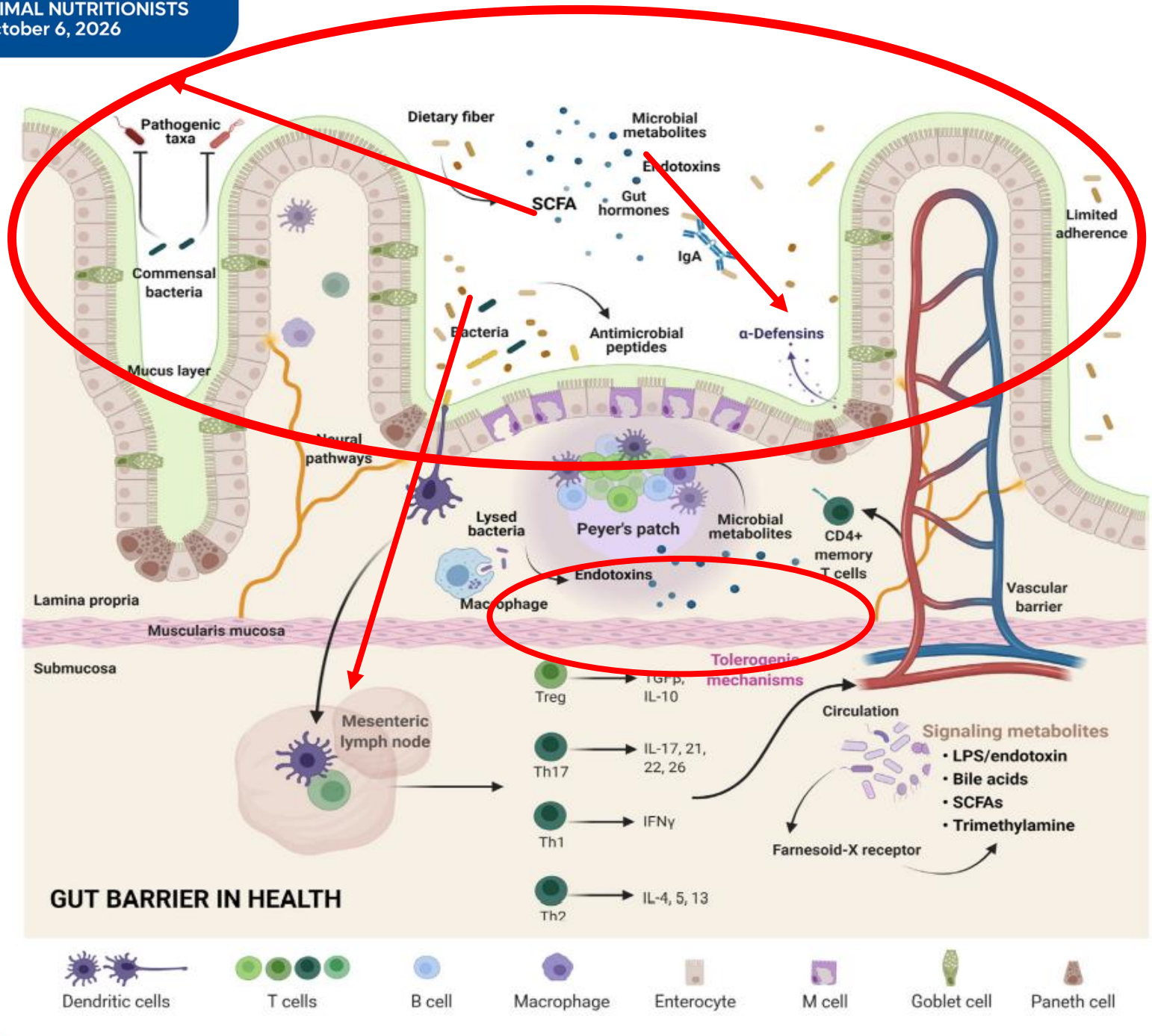
Nutrient utilization → **Barrier integrity** → **Mucosal immunity** → **Microbiota**

These functions are highly interconnected rather than independent.

A disturbance in one component can affect the whole system.

The outcome can be reduced growth, poorer feed efficiency, greater disease susceptibility and welfare challenges.

The Gut Is More Than a Digestive Organ



(J CLIN EXP HEPATOL 2022;12:625–638)

The Gut–Immune Axis

A continuous two-way communication network



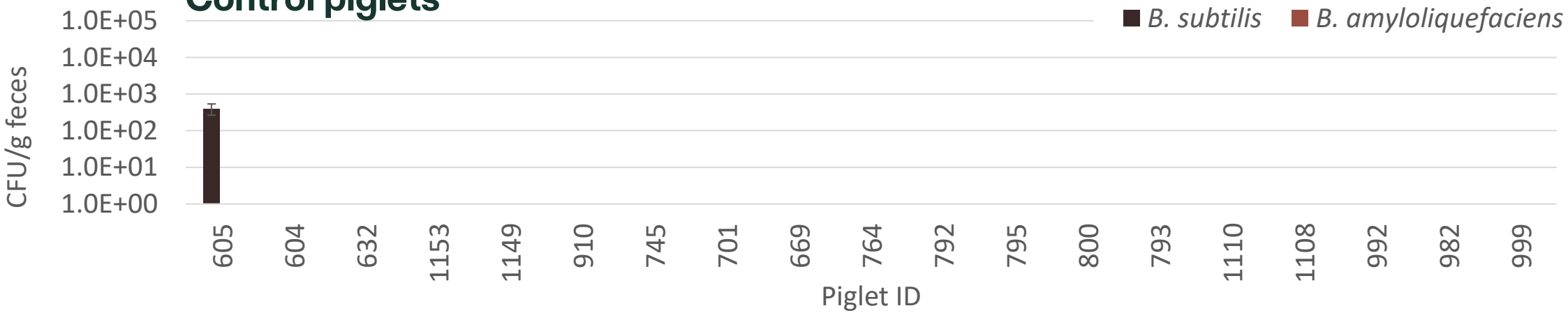
Microbial metabolites and microbial signals influence immune development and epithelial function.

Immune signals, in turn, shape the microbial ecosystem and barrier environment.

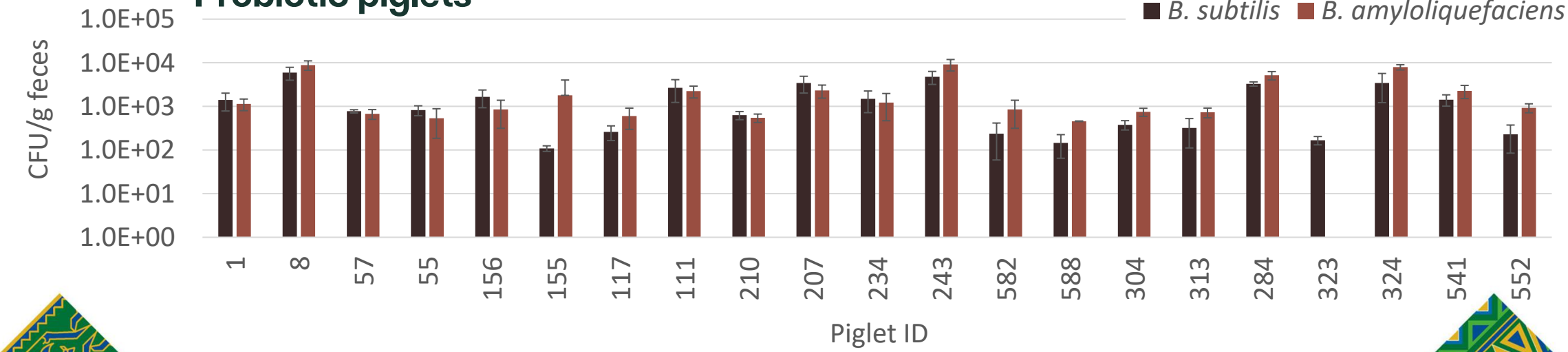
Sow-Piglet connection (natural route)

Maternal transfer of strains to one day old piglets

Control piglets



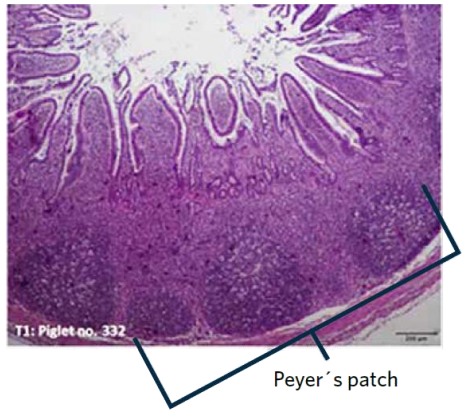
Probiotic piglets



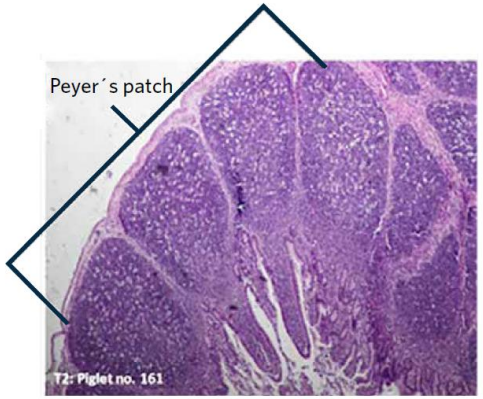
Source: Internal trial ID 80675

Probiotics matures the immune system of piglets at weaning

Without Probiotics



With Probiotics

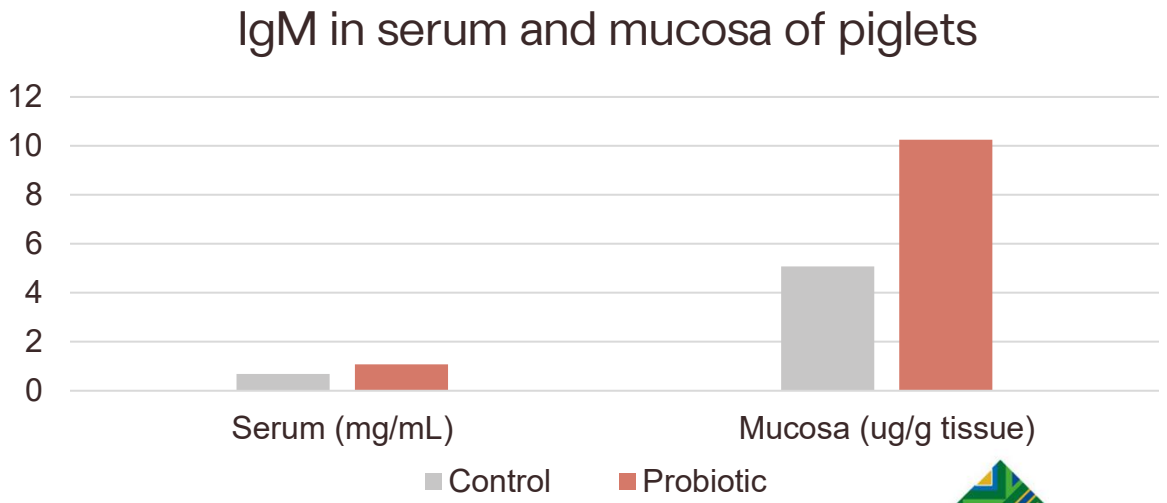


	Control	Probiotic
Mucosa thickness (µm)	443.1[a]	529.0[b]
Average area of Peyer patch (µm^2)	311.9[a]	434.4[b]

Values lacking a common letter are different (P ≤ 0.05)

Source: Internal report 80674

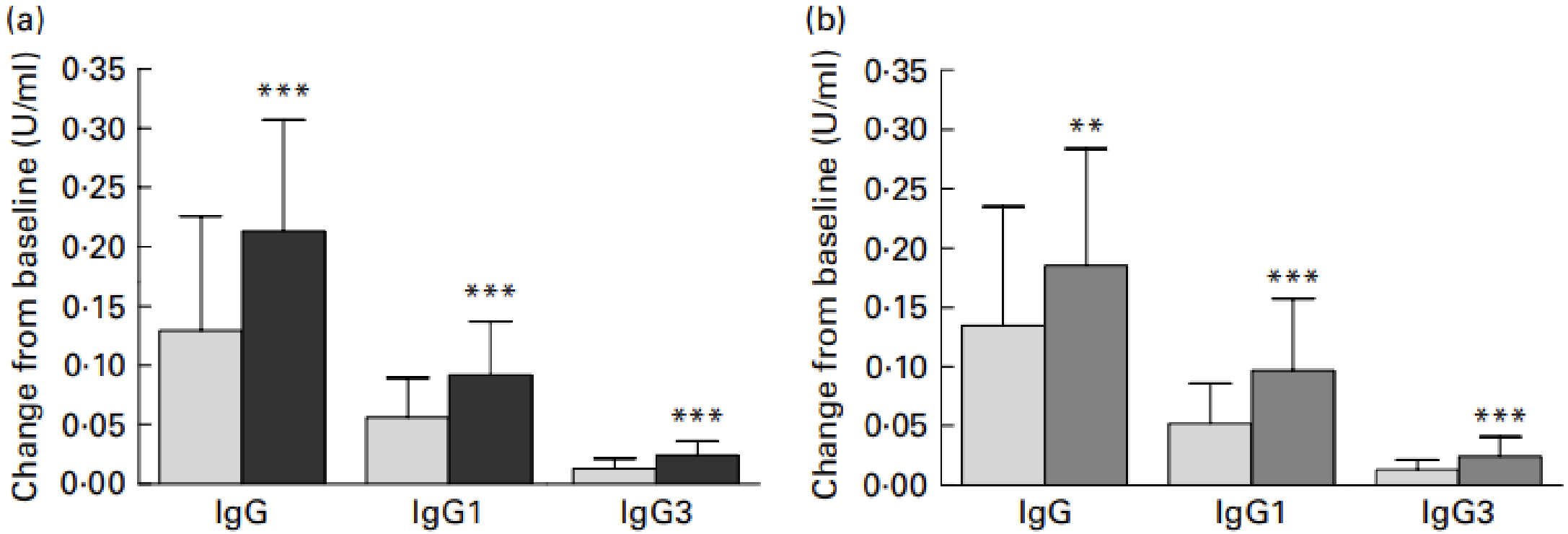
The lymphoid cells of Peyer's patches are predominantly producing the IgM type antibodies



Placebo-controlled study in an influenza human vaccination model

Immune benefits of BB-12® and L. casei 431®

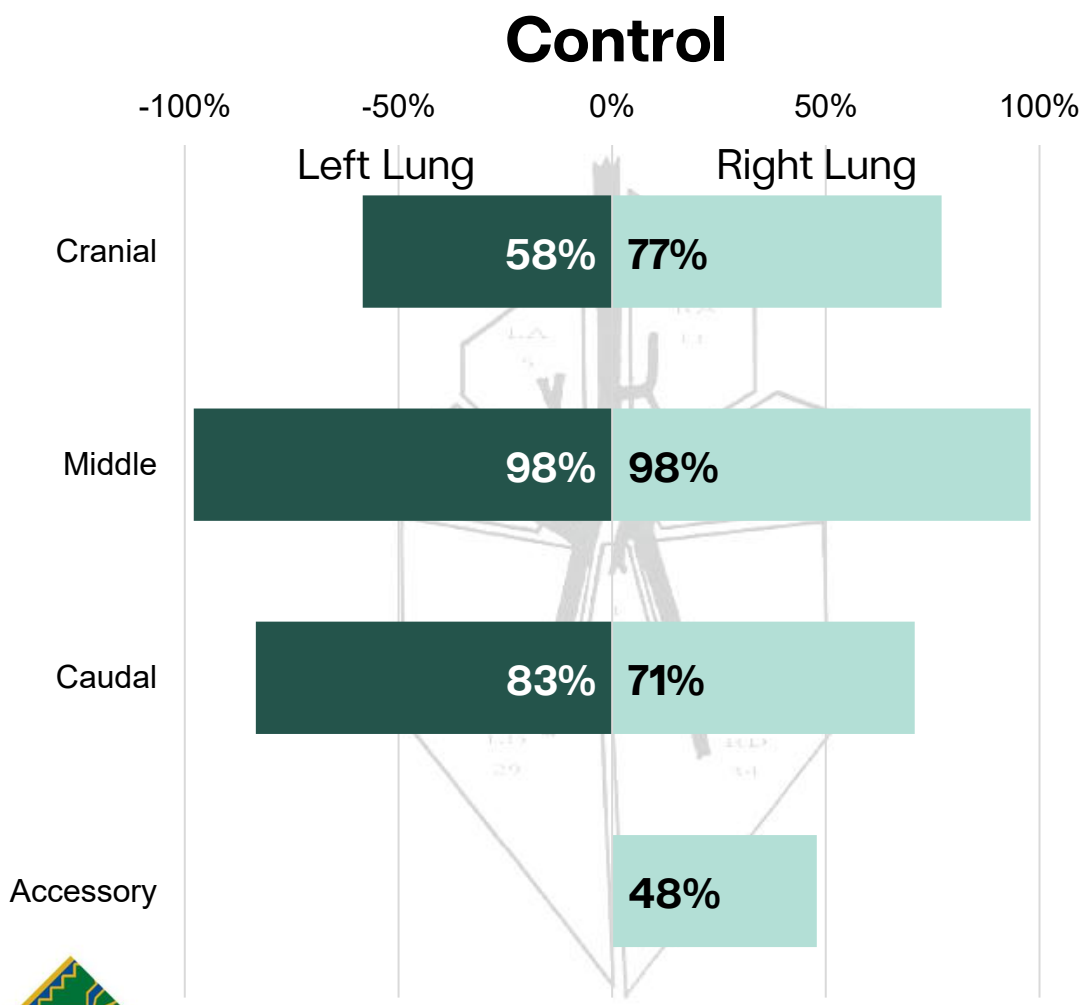
vaccine-specific plasma IgG, IgG1 and IgG3

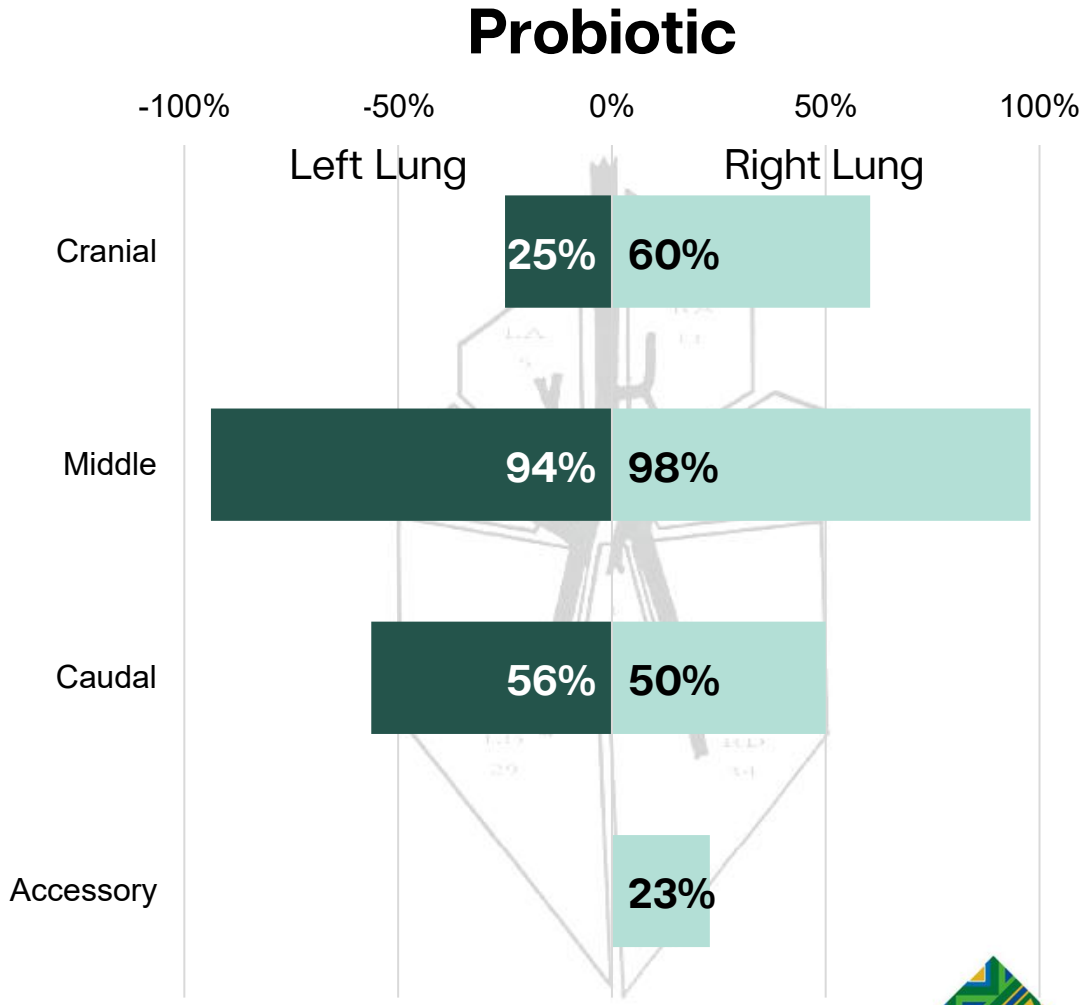


British Journal of Nutrition (2012), 107, 876–884

Influenza proof of concept:

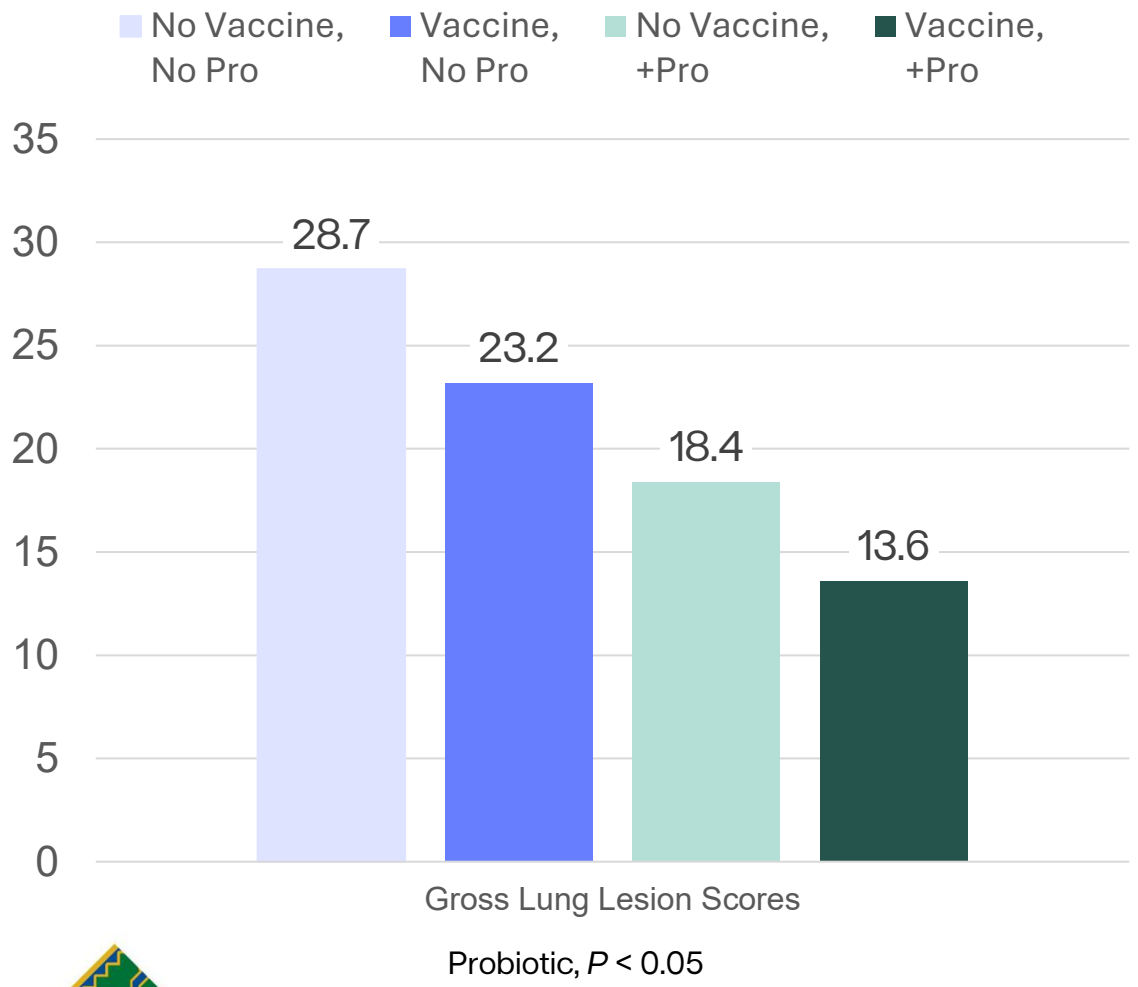
Significantly fewer lung lesions in the probiotic group . . .





X^2 test; $P<0.0001$

Prior experience with viral challenges

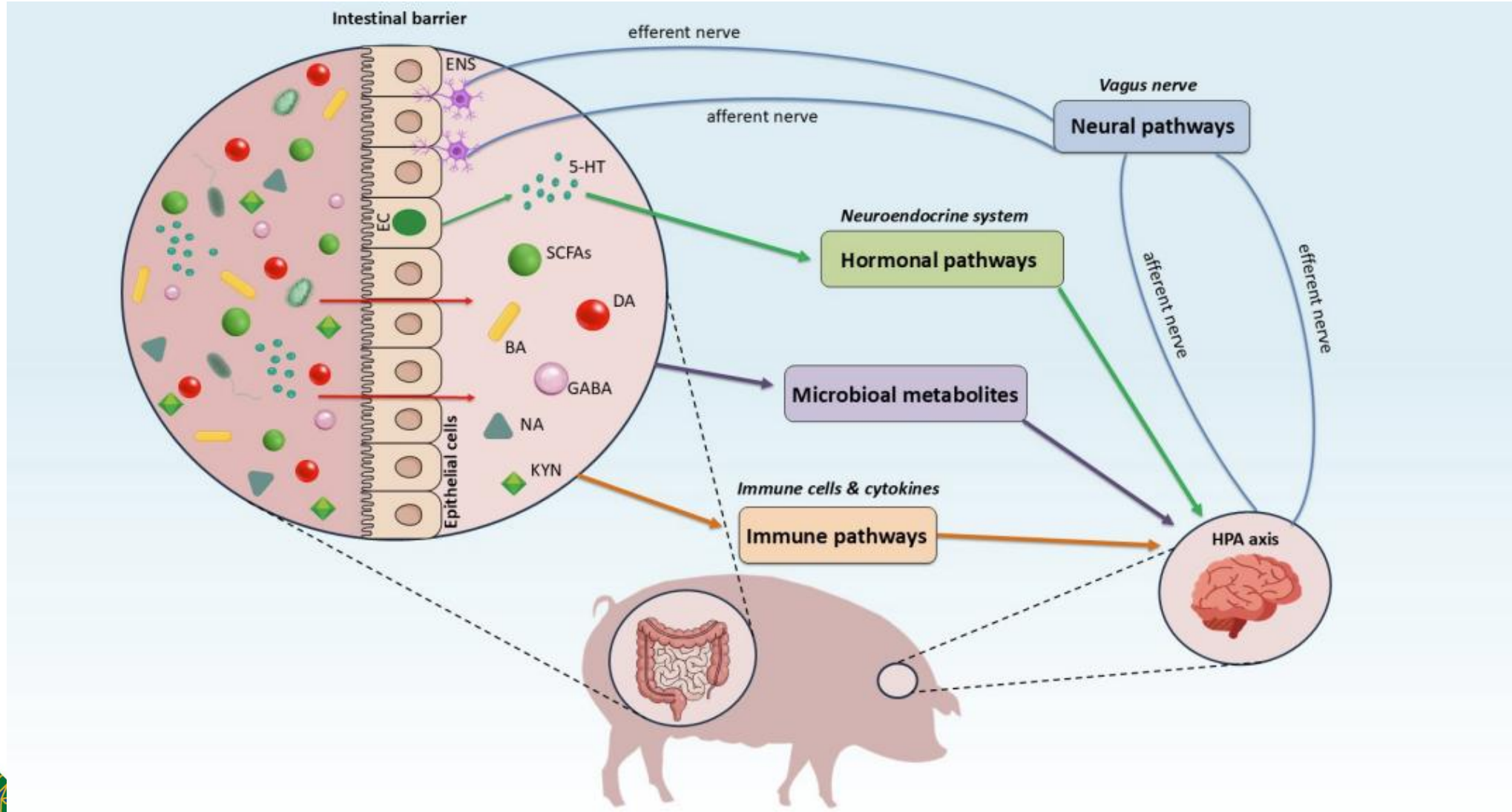


PRRS Challenge Study

- Feeding of an effective probiotic associated with:
  - Maintained weight gain in post-challenge period
  - Greater ADG versus control ($P < 0.06$)
  - Lower F:G versus control ($P < 0.05$)
  - Lower gross lung lesion scores ($P < 0.05$)

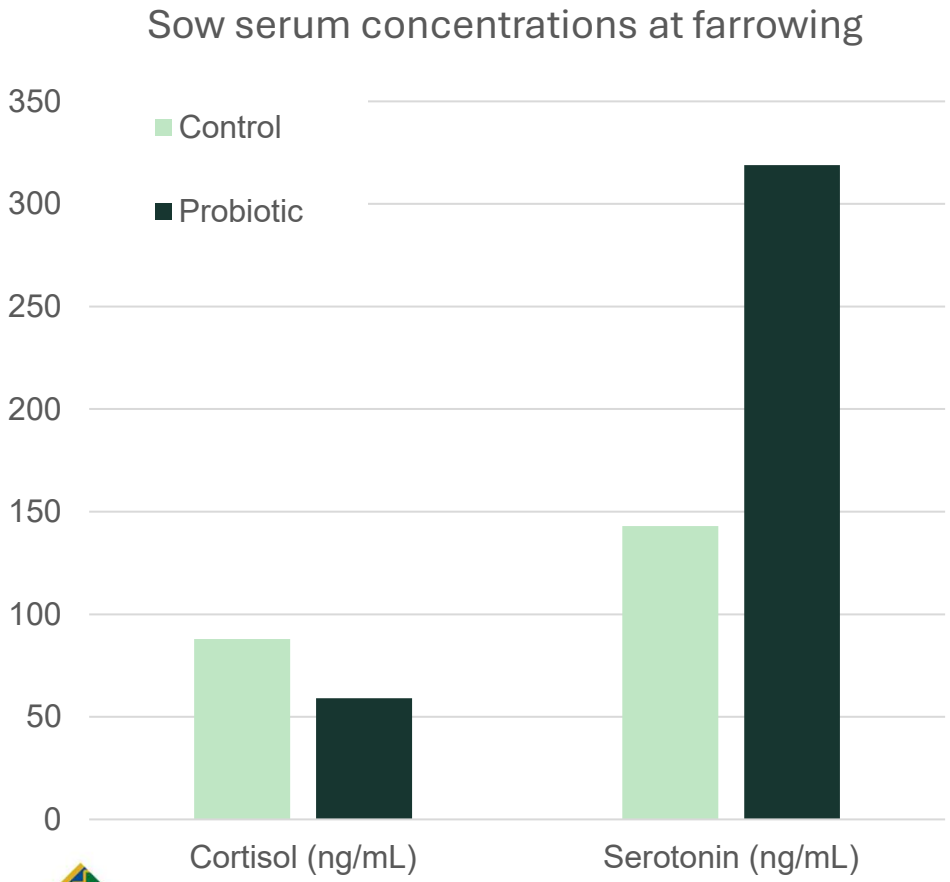
Chr. Hansen Study 80700 and 80707

Gut-Brain Axis

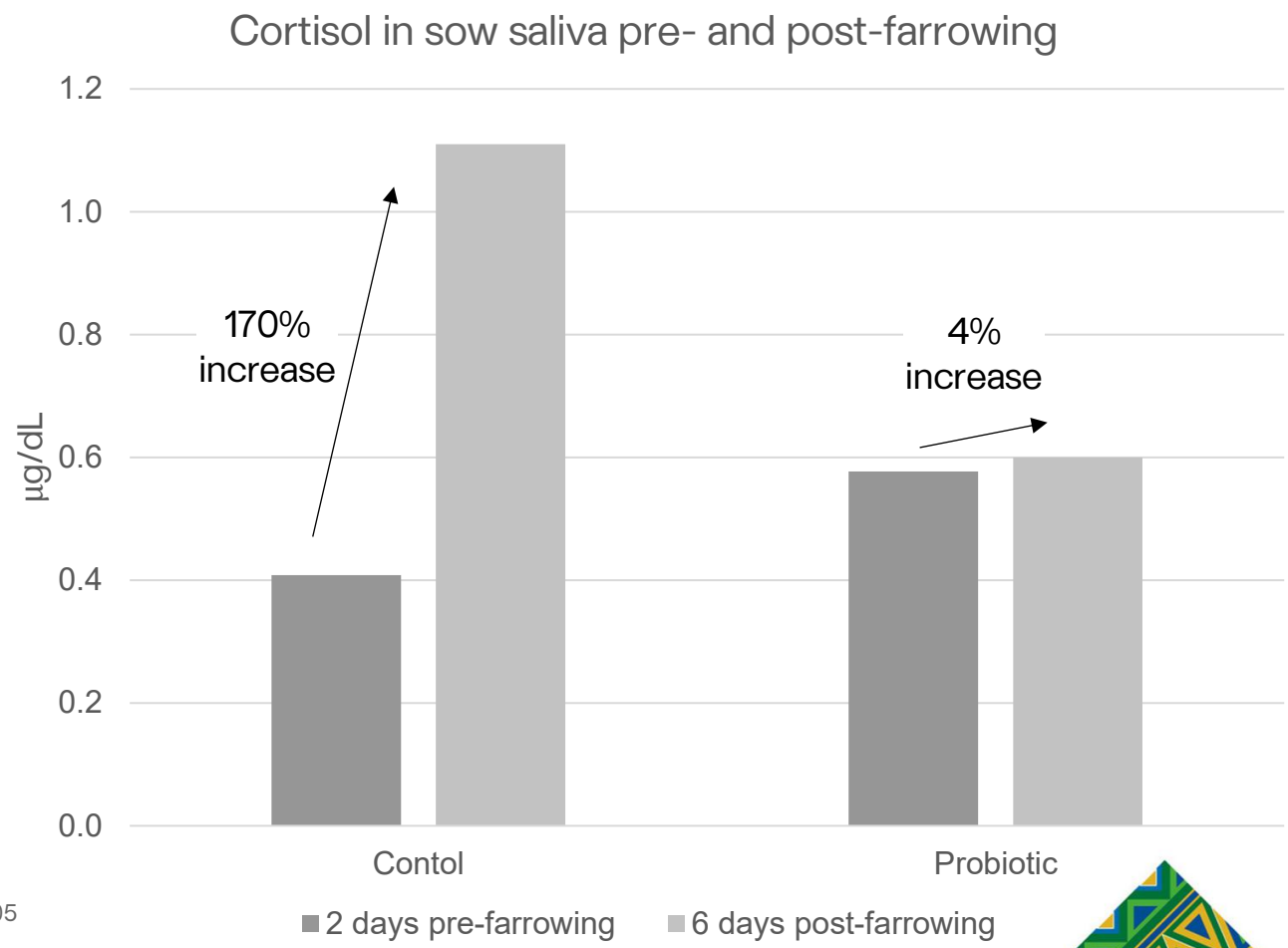


Animals 2025, 15, 3653

Effect of probiotic to stress hormone



All differences between treatment groups are $P < 0.05$



All differences between treatment groups are $P < 0.05$

Source: Internal trial IDs 80928 & 80845

Production Group	N	Duration	Behavioral Group		Microbiota Changes		Study
Finishing pigs (Landrace × Large White hybrid sows sired with Duroc × synthetic hybrid boar)	352	4 weeks	Tail-biting pigs (aggressor)/bitten finisher pigs (victim)	Aggressor	↑*Coprococcus* ↑*Clostridium IV*	↓*Lactobacillus*	[7]
				Victim	↑*Sphaerochaeta* ↑*Blautia*, ↑*Phascolarctobacterium*	↓*Lactobacillus*	
Swine (Swedish Landrace Yorkshire × Hampshire)	29	2 years	Tail-biting piglets (aggressor)/bitten piglets (victim)	Aggressor	↑*Firmicutes* (mainly *Lachnospiraceae*, *Ruminococcaceae*)		[50]
				Victim	↑*Firmicutes* (mainly *LachnospiraceaeiButyrivibrio*), ↑ *Bacteroidetes* (*Alloprevotella*)	↓*Bacteroidetes* (mainly ↓*Prevotella 7*), ↓*Proteobacteria* (*Ralstonia*)	
Piglets (Finnish Landrace × Yorkshire × Duroc)	30	Single-day study (day 45)	Piglets exhibiting manipulative behavior (manipulators)	Manipulators	↑ *Lactobacillus amylovorus*	↓ diversity of *Lactobacillaceae*	[51]
Suckling piglets (2-4 tyg.) Topigs-20	47	4 weeks	Piglets exhibiting increased/decreased exploratory activity	↑ Exploratory activity	↑*Atopobium* ↑*Coprococcus 3* ↑*CAG-873* ↑*Eubacterium* ↑*Coprostanoligenes*		[13]
				↓ Exploratory activity	↑*Prevotella 9*		

N—number of animals; Duration—duration of experiment; ↑ Increase; ↓ Decrease.



Summary

- The Gut Is More Than a Digestive Organ
- Microbiome plays an important role in swine health
- Sow microbiome is associated with the reproductive cycle and linked to productivity
- Effective biosolutions generally have a significant modulation of the microbiome
- Effective biosolutions modulate sow and piglet microbiome and is associated with health and production.

thank you

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